DHCP

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ITT-121

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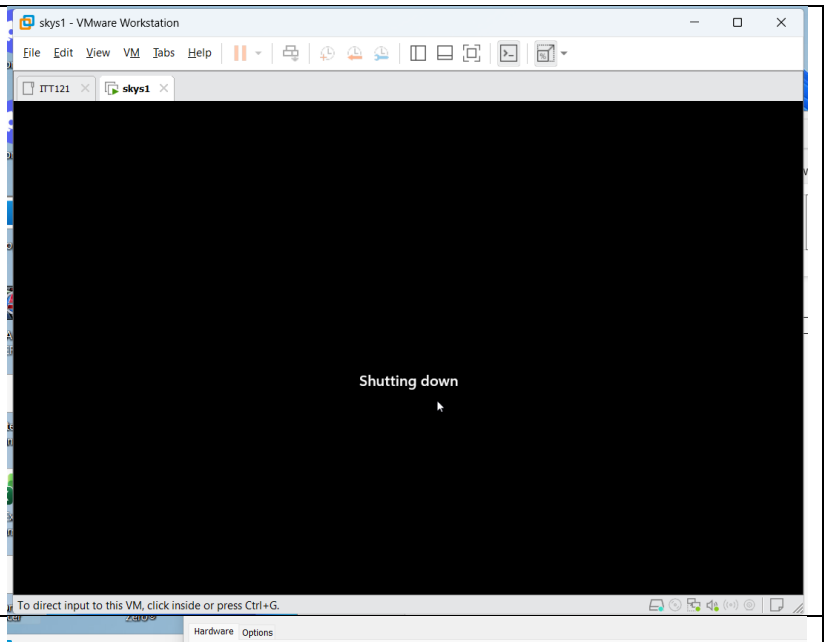
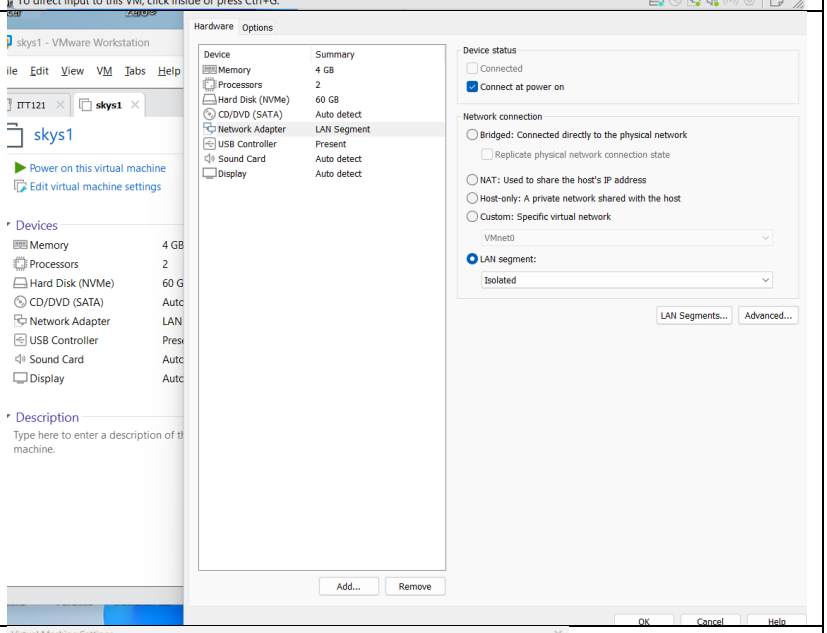
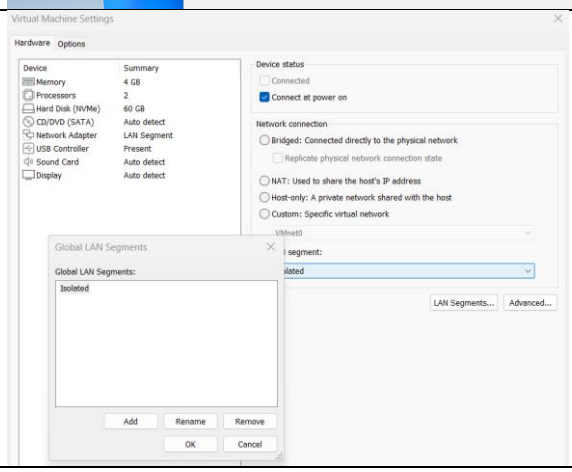
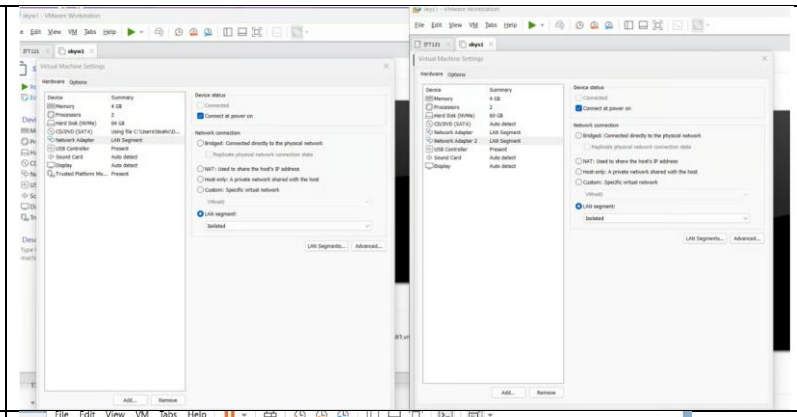
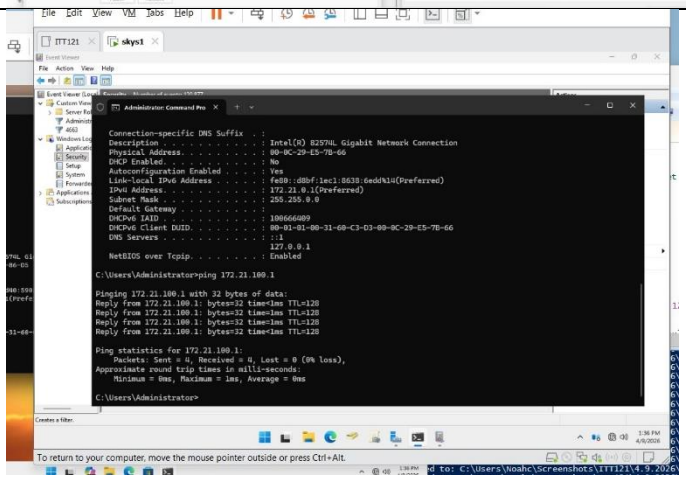
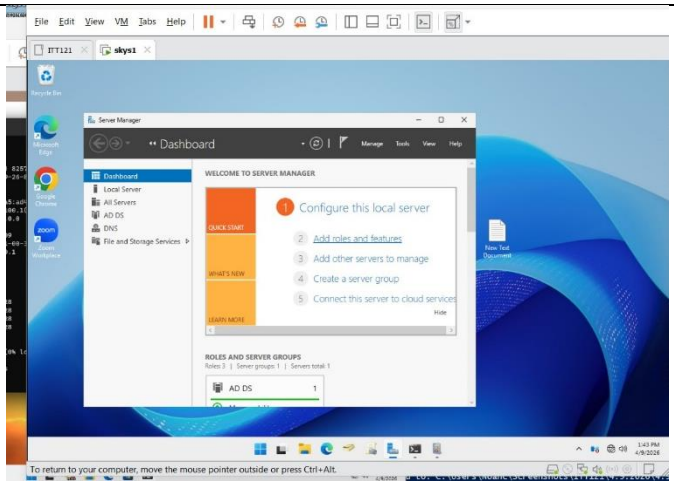
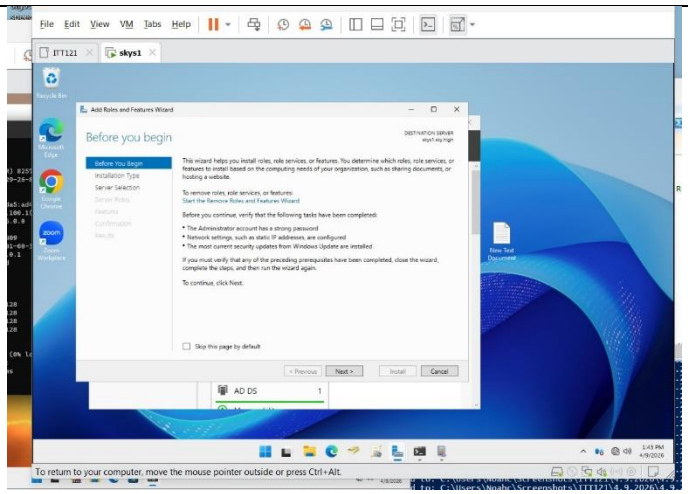
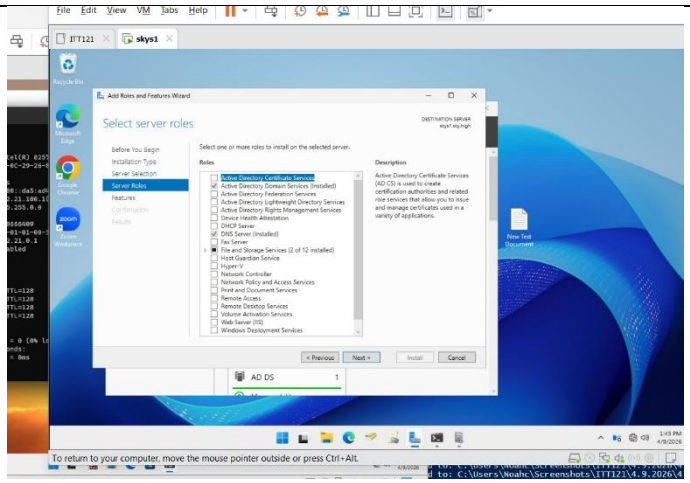
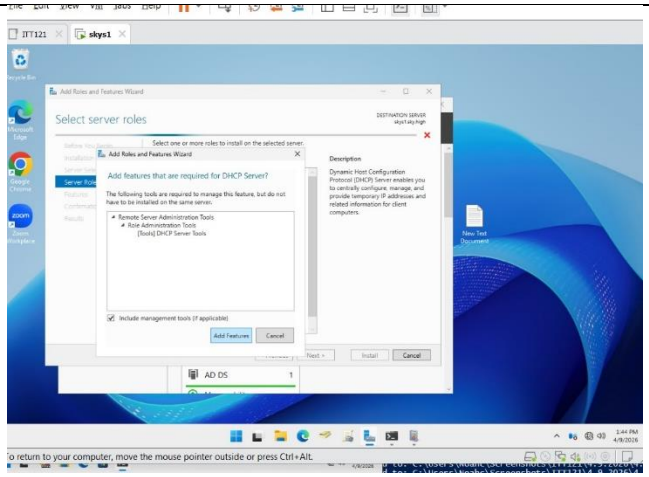
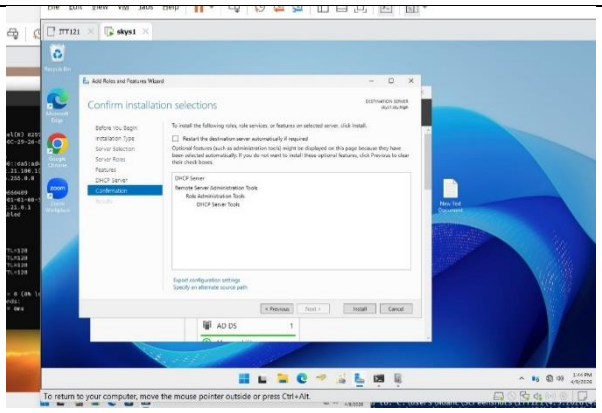
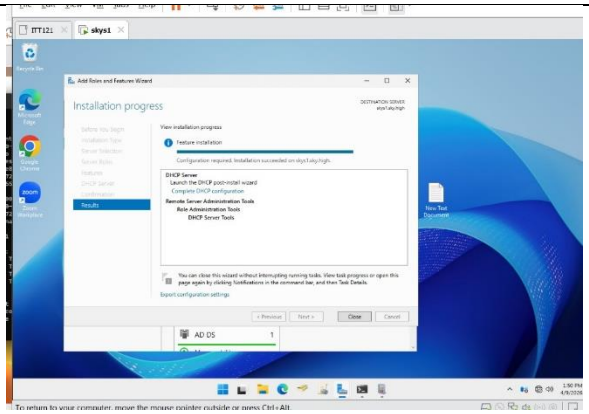
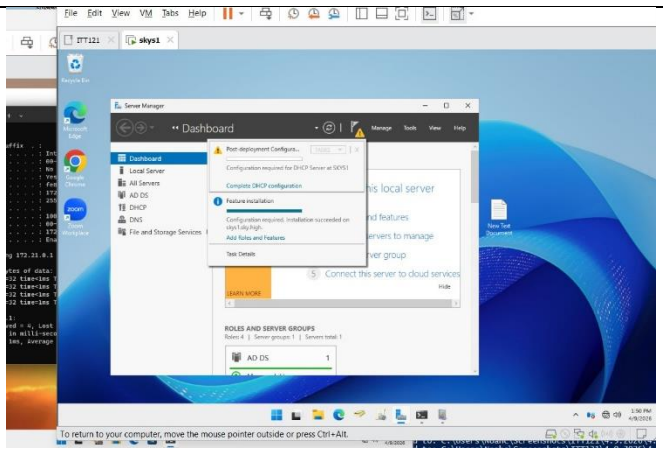
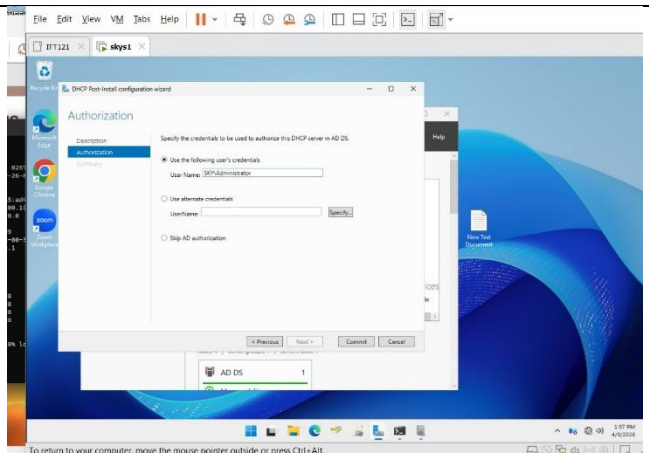
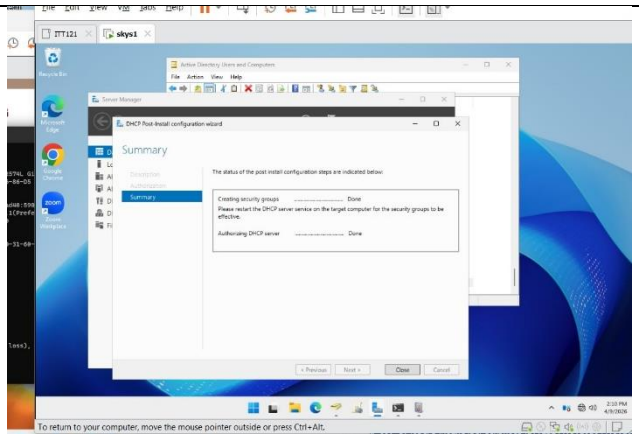
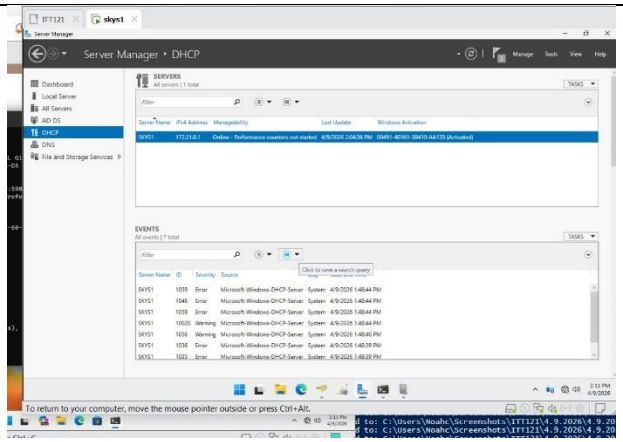
April 11, 2026

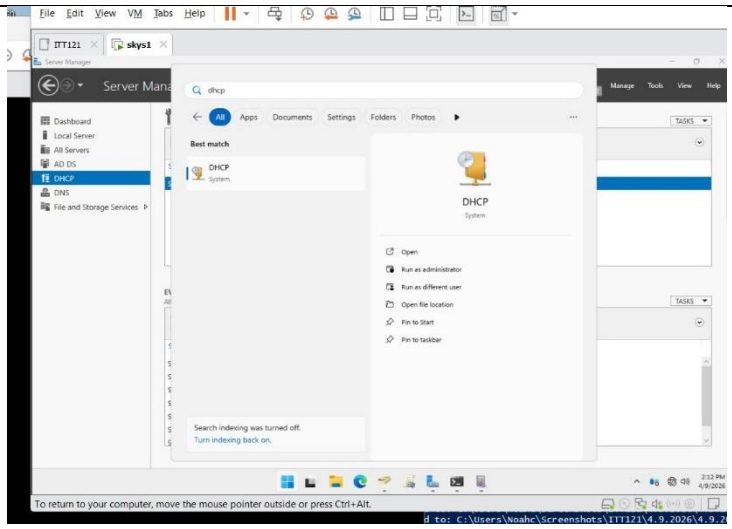
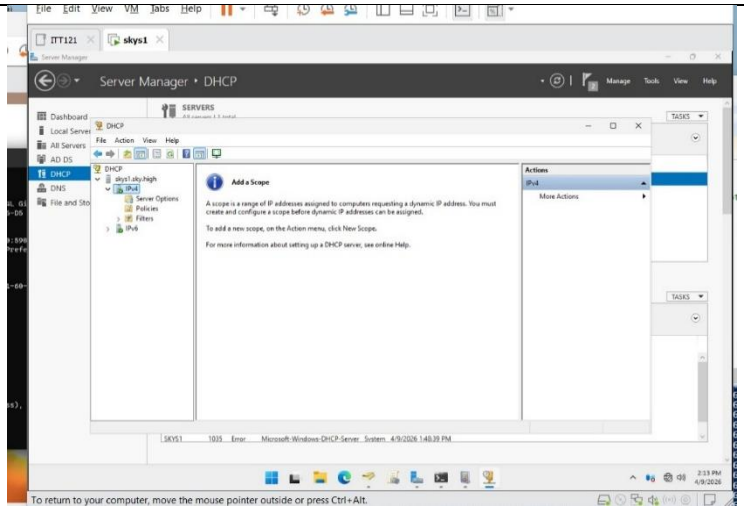
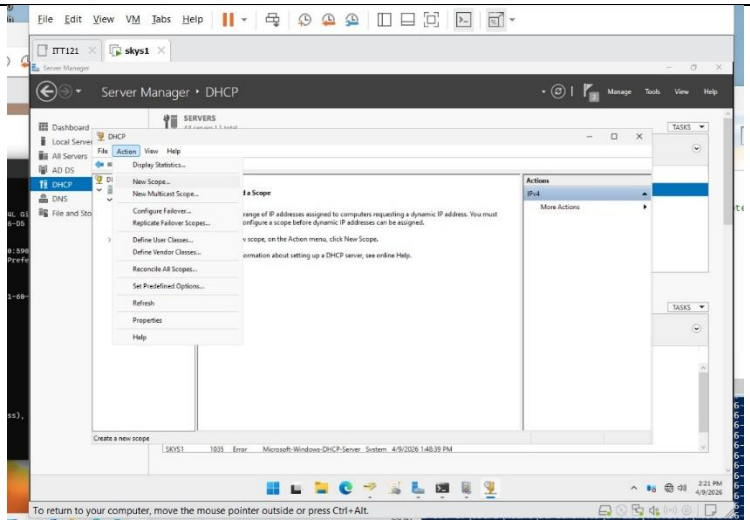
Before we add DHCP to our server we need to isolate our network so our DHCP server does not get out of our machines. To do this we need to shut down the server and the workstation.	
When the machines are off click on edit virtual machine setting and find the network adapter. Check the box "LAN segments" and then click on the box "LAN Segments…"	
Add a new segments and give it name like Isolated.	

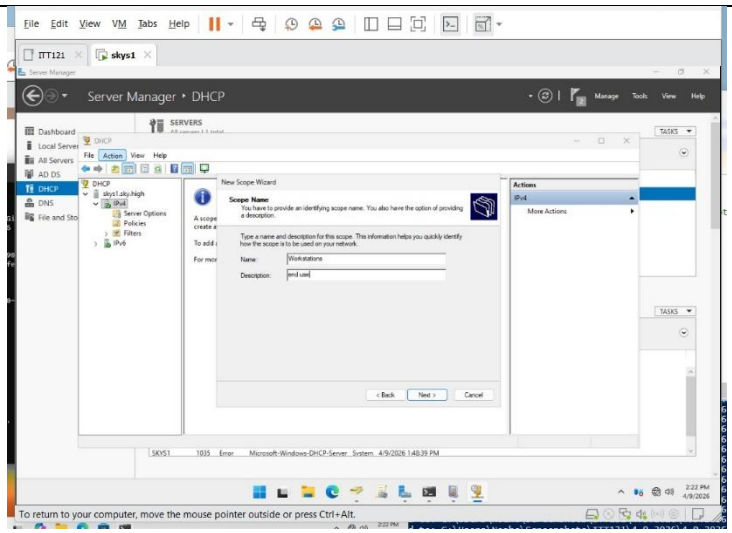
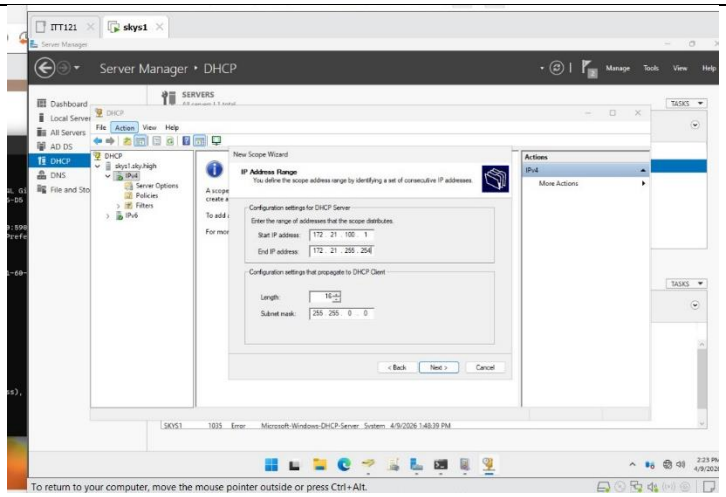
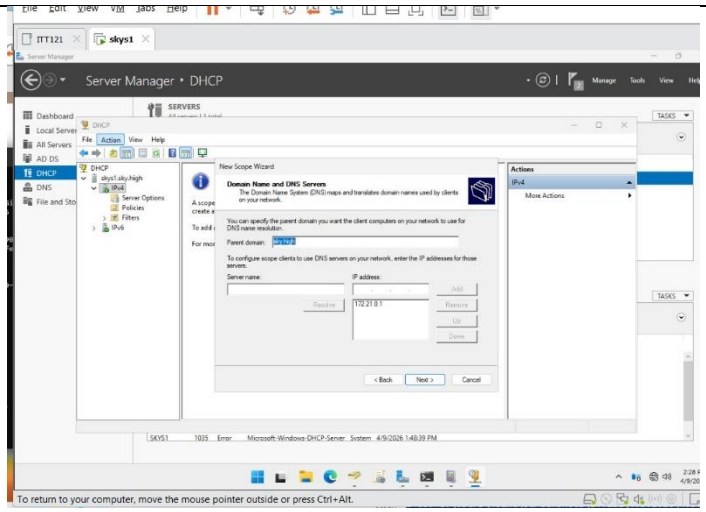
For both of the machines make sure that the network adapter is using a LAN Segment and make sure that Isolated is selected for both machines.	
Now start the workstation and server again and test the connection between the two machines by pinging from command prompt. To open command prompt click the windows icon and type in “cmd” and then hit enter on the keyboard.	
Now in the server bring up the Server Manager and click of “Add roles and features”.	

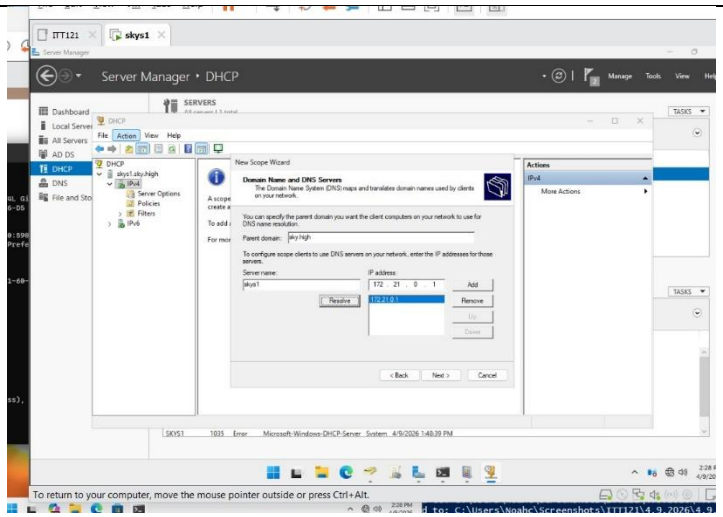
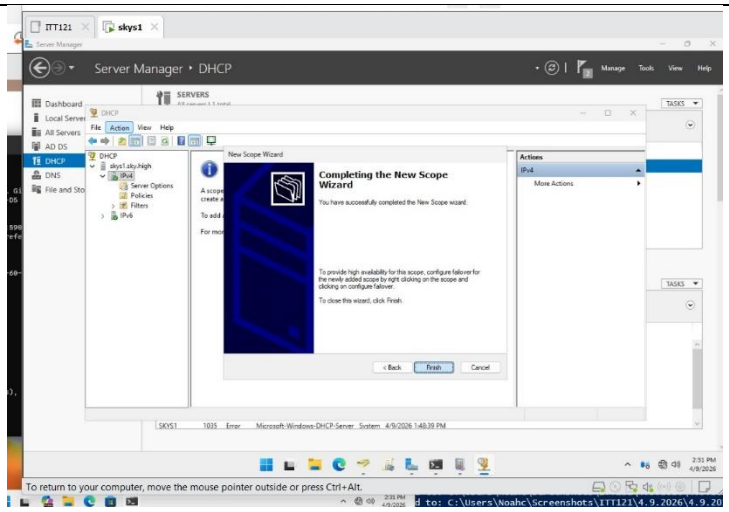
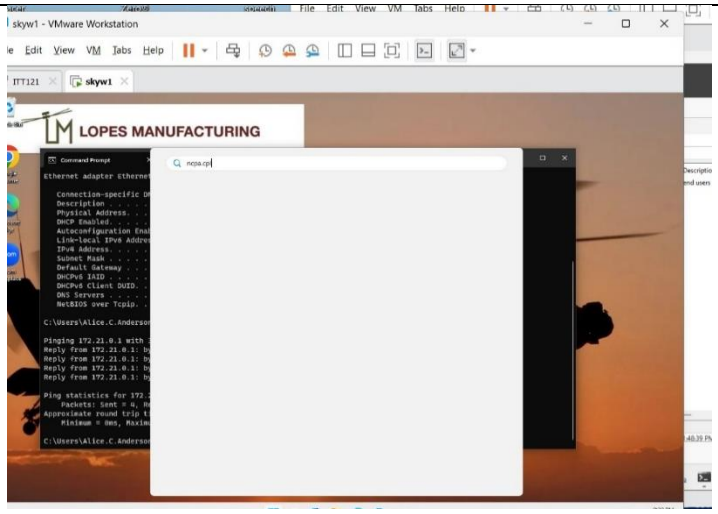
Continue through the pages by hitting next	
Stop at Select Server Roles and then find and check the DHCP option.	
When you hit next youll see a more detailed list of what you are installing, you can click Add Features and continue.	

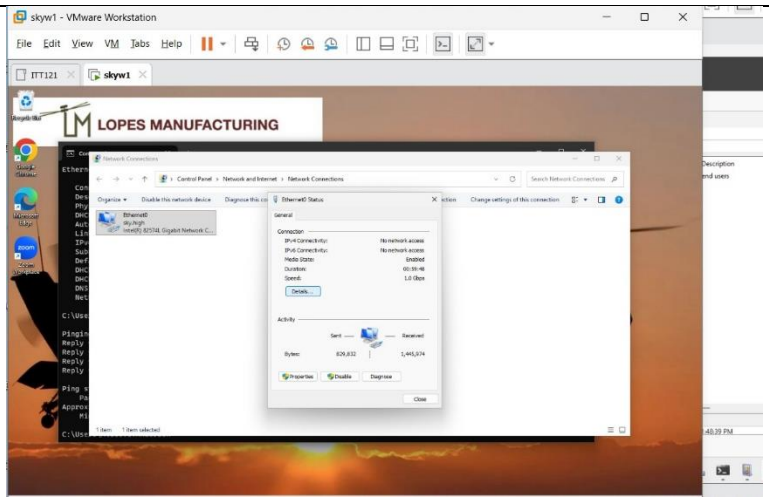
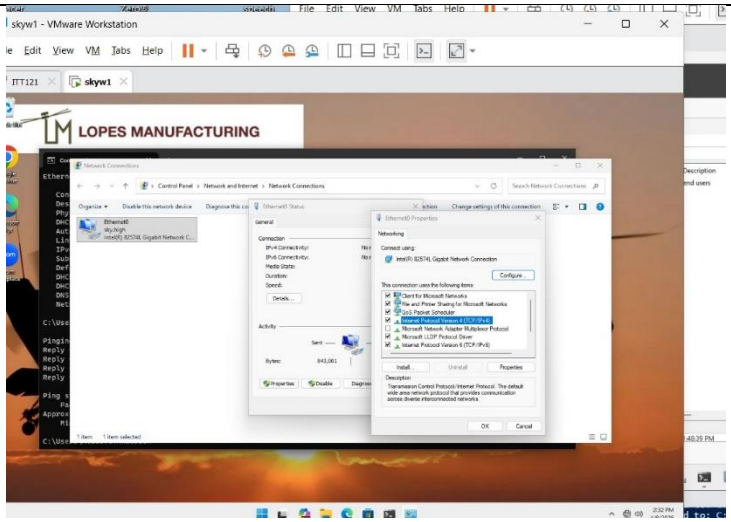
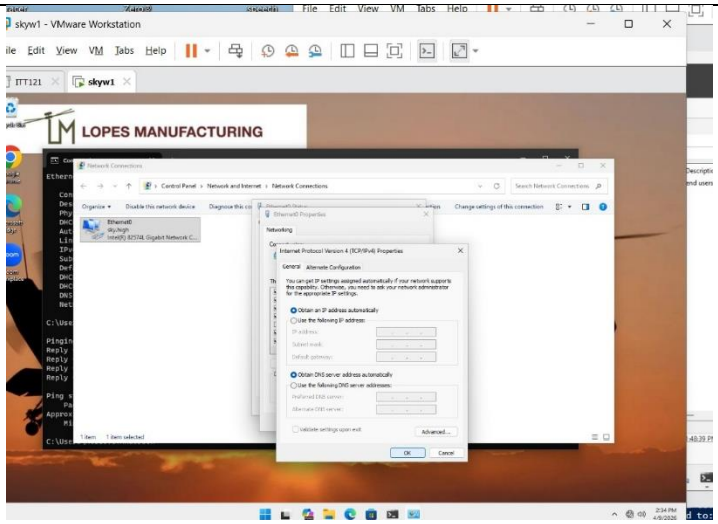
Keep clicking next on the pages until you get the Confirm Installation Selections page. Click install.	
Once the install is done click close.	
Now in the Server Manager there is a flag in the top toolbar with a caution icon. Click on the flag and the click “Complete DHCP Configuration”	

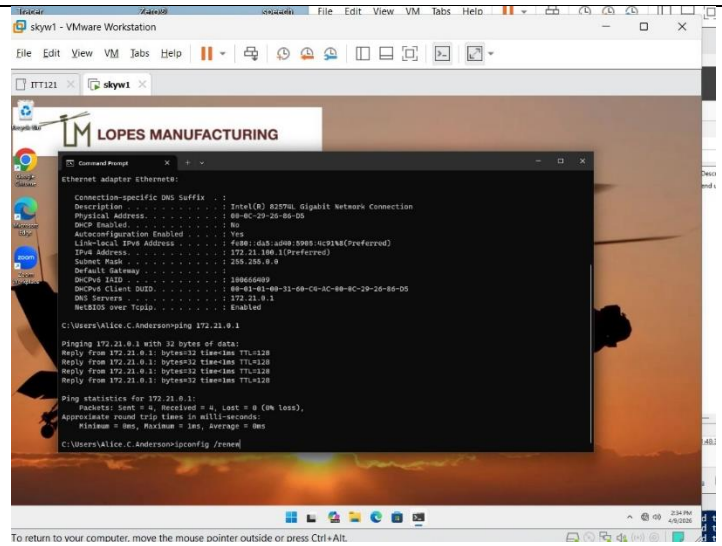
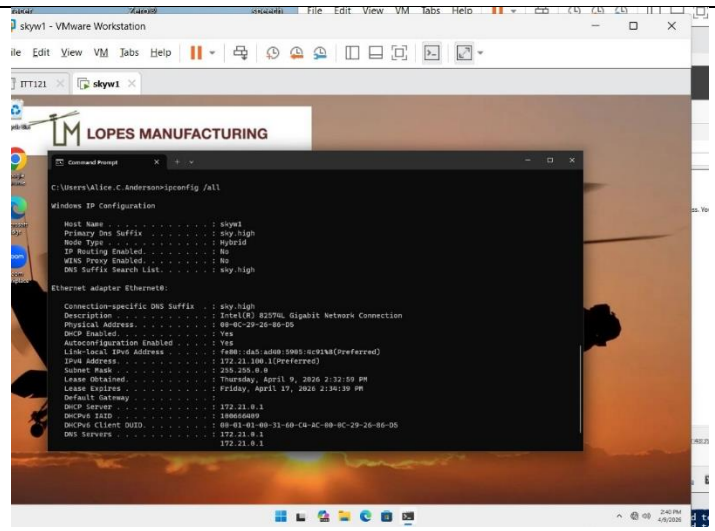
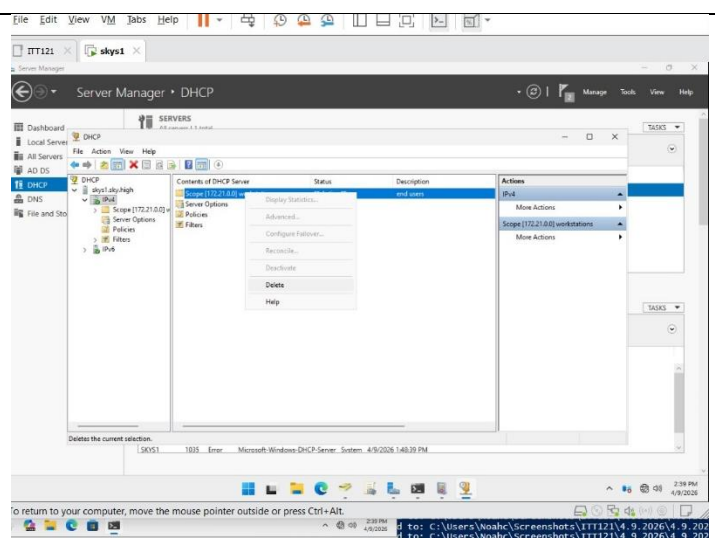
After the description we will have the option to specify the credential to authorize. It would be best practice to not have this as the main Administrator, but with how our current server and workstation are setup we can use the Administrator this time. Click the commit button.	
In the summary both tasks should be marked with Done. Click close.	
Now in the server manager go to DHCP by clicking on it on the left hand side. You should see that the server is activated under the Windows Activation side.	

Now we need to add a scope. Click the windows icon and type in DHCP and then hit enter.	
In the DHCP application expand the domain name and find the IPv4 option.	
Right click IPv4 and click New Scope.	

Give the scope a descriptive name like “workstations” and a description of “end users” so we know what this scope is for.	
For the IP Address Range the start IP address is 172.#.100.1 The end IP address is 172.#.255.254 Where # is the number you were assigned at the start of the semester. Length is 16 and the subnet mask is 255.255.0.0 Hit next	
Continue through the pages until you are at the Domain Name and DNS Servers page. Here we need to add our dns server.	

Type in your server name in the server name box and hit Resolve to automatically fill out the IP address or enter the IP address manually. Click add and then next.	
Now we can finish the New Scope setup. Click Finish.	
Now switch to the workstation and hit the windows icon to open the search bar. Type out ncpa.cpl and then hit enter.	

Click on Ethernet when the application opens, and then click on properties.	
In the list double click IPv4	
For IP address and DNS server click the box “Obtain IP/DNS automatically”. Hit ok and then close the applications.	

Now in the workstation open command prompt by clicking the windows icon, tying cmd in the search bar and then hitting enter on the keyboard. In the terminal type “ipconfig /renew” and then hit enter.	
Now to see our new IP address details, type out “ipconfig /all” in the terminal and hit enter. In the details we can see that DHCP is enabled, and that autoconfiguration is enabled.	
For this class, we don’t want to keep DHCP around because if DHCP gets out of our machines we can take out other devices on the network. So in the server, go back to DHCP and delete the scope we just made. Another option is to revert back to a snapshot before we installed DHCP on the server.	

DHCP plays a vital role in minimizing administrative workload within Lopes Manufacturing by automatically assigning and managing IP addresses for numerous devices, including employee workstations and technicians' laptops. Without this automation, IT staff would have to manually configure network settings on each device, which increases the likelihood of errors, duplicate IP addresses, and inefficient use of time in a high-demand environment that supports critical military operations. By assigning static IP addresses to essential infrastructure such as servers, printers, and domain controllers, and using DHCP to distribute dynamic addresses to user devices, the company maintains both reliability and flexibility. Additionally, DHCP streamlines the process of adding new devices, allows the network to scale more easily, and enables centralized control over key settings like DNS and default gateways. This reduces routine administrative tasks and allows IT personnel to prioritize security and system performance, ensuring consistent and dependable network functionality for mission-critical operations.

References

Dynamic host configuration protocol (DHCP). GeeksforGeeks. (n.d.) https://www.geeksforgeeks.org/computer-networks/dynamic-host-configuration-protocol-dhcp/

Dynamic host configuration protocol (DHCP) overview. Microsoft Learn. (n.d.). https://learn.microsoft.com/en-us/windows-server/networking/technologies/dhcp/dhcp-top